

CHM 3120: Introduction to Analytical Chemistry

Unit 2 - Section 3:

Quality Assurance & Figures of Merit for Analytical Methods

Chapter 6

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Quality Assurance and Calibrations

What is quality assurance (QA)?

How do we compare analytical methods?

What constitutes signal and what constitutes noise?

How do we pose a scientific question?

Basics of Quality Assurance

False positives vs false negatives

Suppose 0.2% of people have a particular type of cancer and an analytical test has a 99% probability of detecting cancer with a 1% false positive rate.

Amongst 1M people, 0.2% (2,000) then likely have cancer.

If 1M are screen, the test will accurately determine that 99% of those 2,000 (1,980) have cancer. But 1% (20) will be incorrectly told they are cancer free (false negative).

In the remaining 998,000 people, a 1% false positive rate identifies cancer in 9,980 healthy people. So, of the positive results, only 1,980/9,980 (17%) are true positives. The remaining 83% are false positives.

Basics of Quality Assurance

False positives vs false negatives

Acceptable false positive and false negative rates **are application dependent**

Ex. You are testing that a contaminant in drinking water is below a legal limit:

- A false positive says that the concentration exceeds the legal limit when in fact the concentration is below the legal limit
- A false negative says that the concentration is below the limit when it is actually above the limit

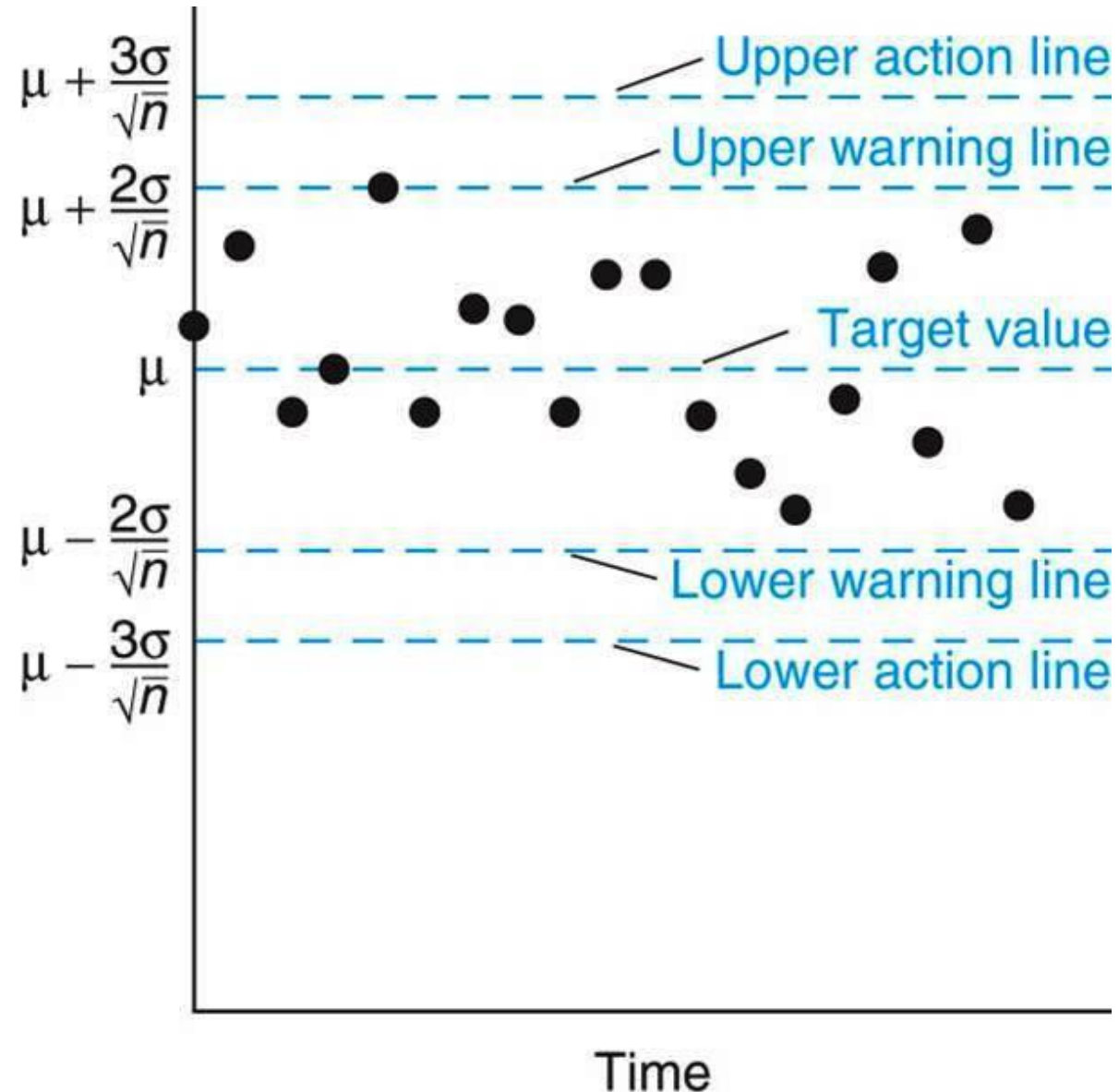
A low rate of false negatives is more important than a low rate of false positives

It would be worse to certify that contaminated water is safe than to certify that safe water is contaminated

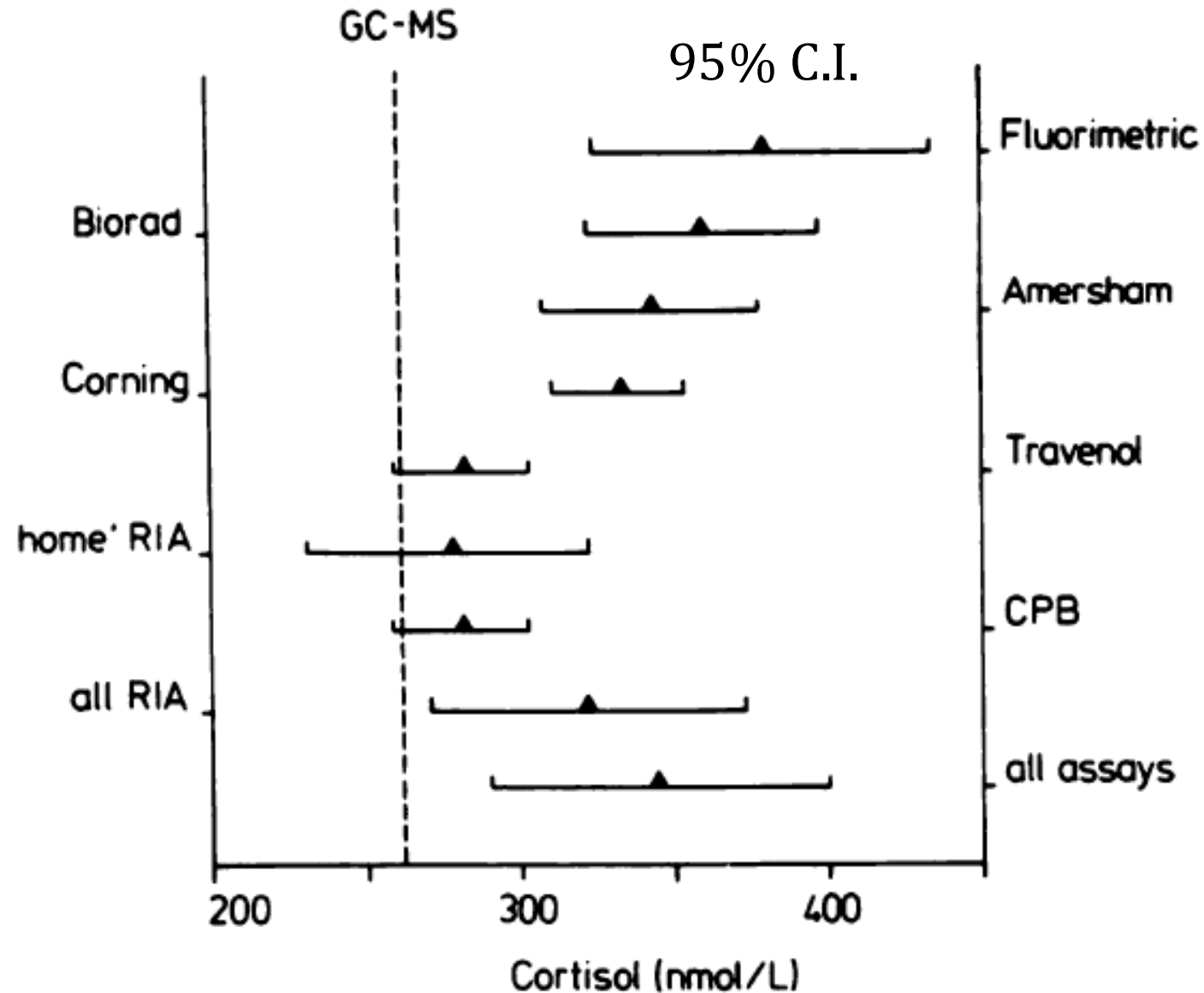
Quality Assurance is Challenging

Vitamin C concentration in
a tablet manufacturing line

~ 95.5%
99.7%
C.I.



Quality Assurance is Challenging



Definitions and Figures of Merit

- **Selectivity or Specificity:** distinguishing analyte from non-analyte
- **Limit of Detection:** a statistical estimate of the lowest detectable amount or conc.
- **Sensitivity:** slope of the calibration function
- **Linearity:** how close a calibration function is to a linear
- **Method Blank:** contains everything but the analyte
- **Field Blank:** a method blank which has been exposed to the sample conditions
- **Matrix:** everything in the sample but the analyte
- **Spike:** a quantity of analyte which is added to the sample
- **Assessment:** data collection to verify analytical performance and objectives
- **Method Validation:** a formal procedure to prove a method is acceptable

Standardization and Calibration

Calibration determines the **relationship between the analytical response and the analyte concentration**, which is usually determined by the use of chemical standards.

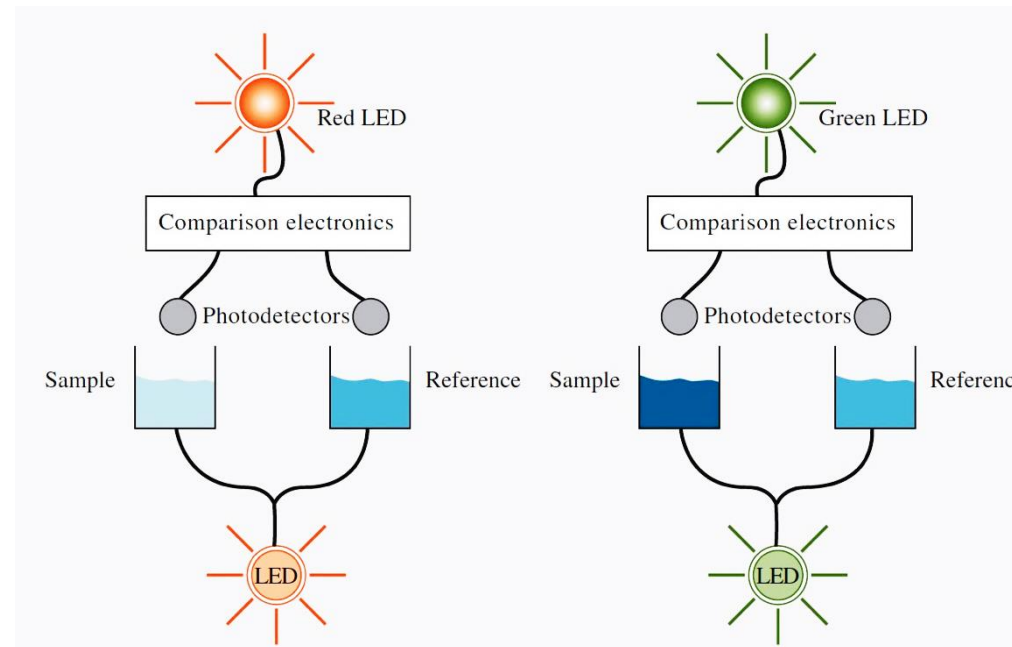
- In the **direct comparison** method (least common), the standard is compared directly to the analyte.
- More commonly, **external standard methods** are used, where standards are prepared separately from the analyte solution.
- When matrix effects from other sample components (called concomitants) may interfere, we can minimize them by:
 - Adding standards into the analyte solution (**standard addition or internal standard methods**), or
 - Modifying or **matching the matrix** composition of standards and samples.

Direct Comparison (null comparison)

This method involves comparing a property of the analyte with that of a standard, adjusting the standard until the measured property matches or nearly matches, producing a null difference.

It gives only an approximate concentration, typically useful near a threshold level.

Most effective when the analyte and standard behave identically. Common in colorimetry, visual titrations, or when using absorbance balancing techniques.



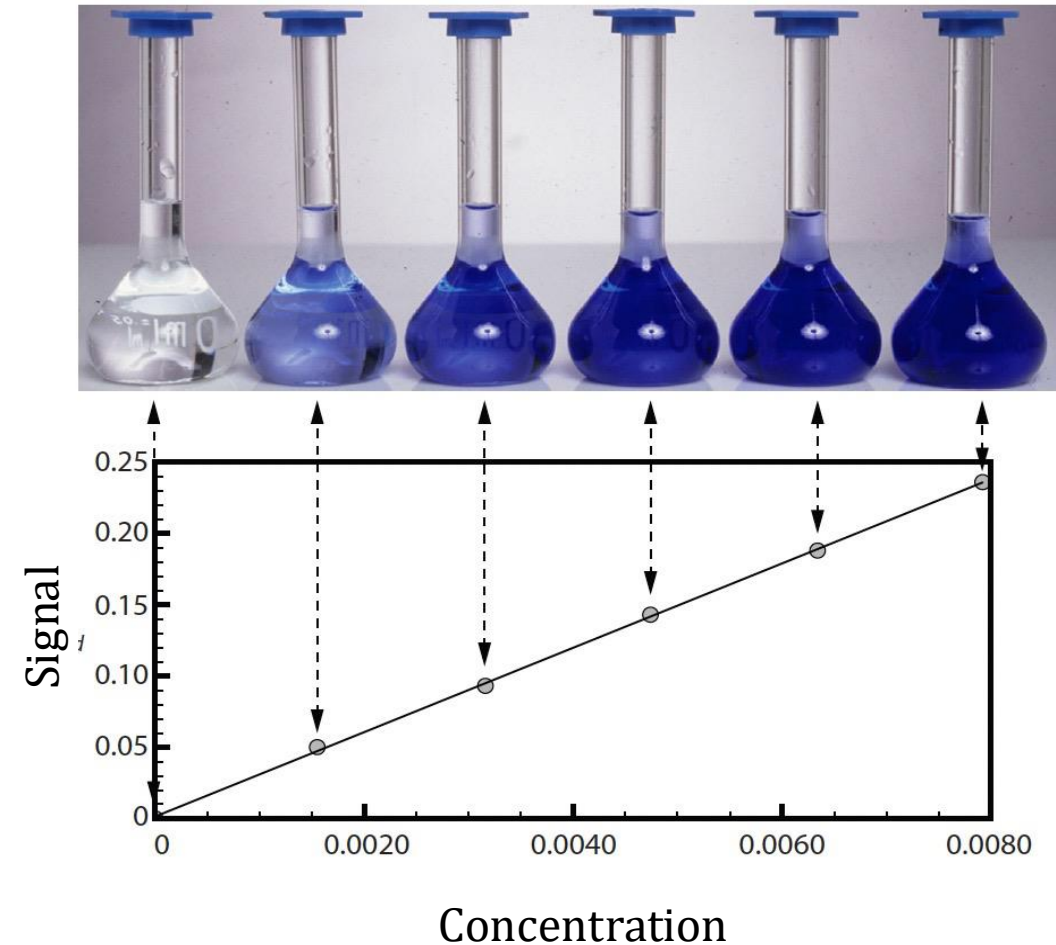
A Comparison Method for Aflatoxins

External Standard Calibration

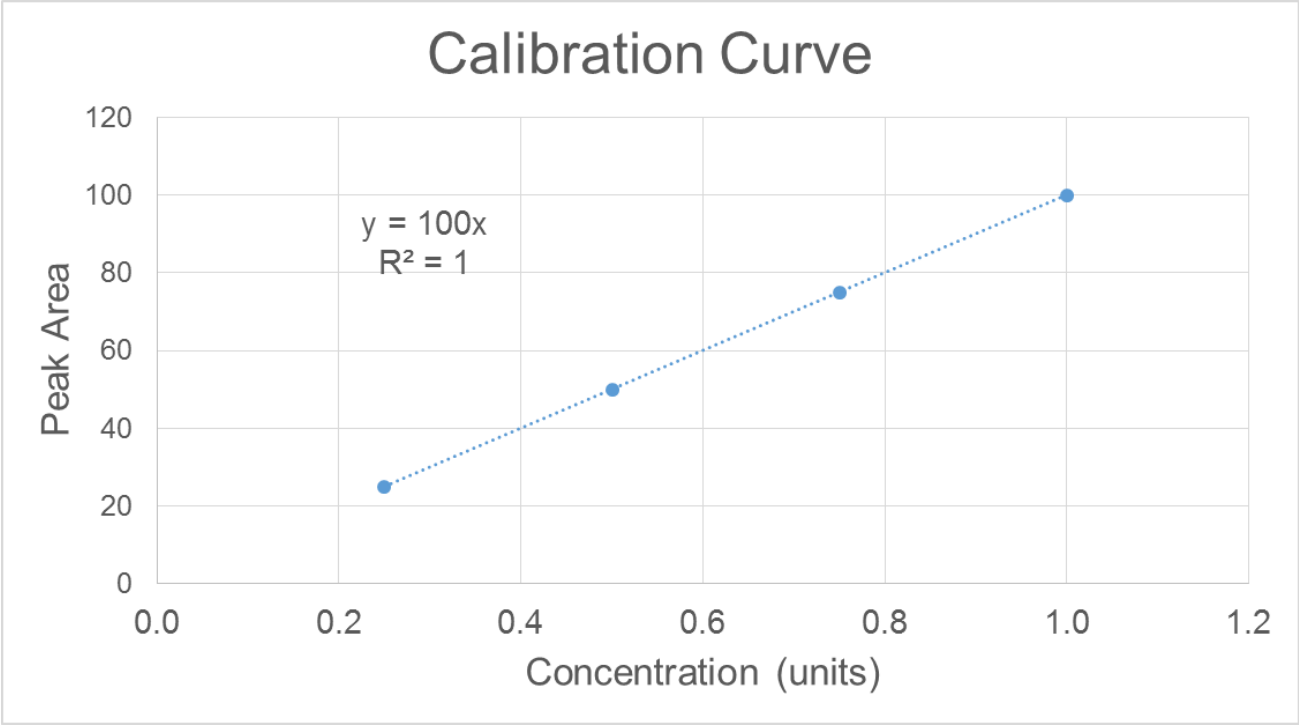
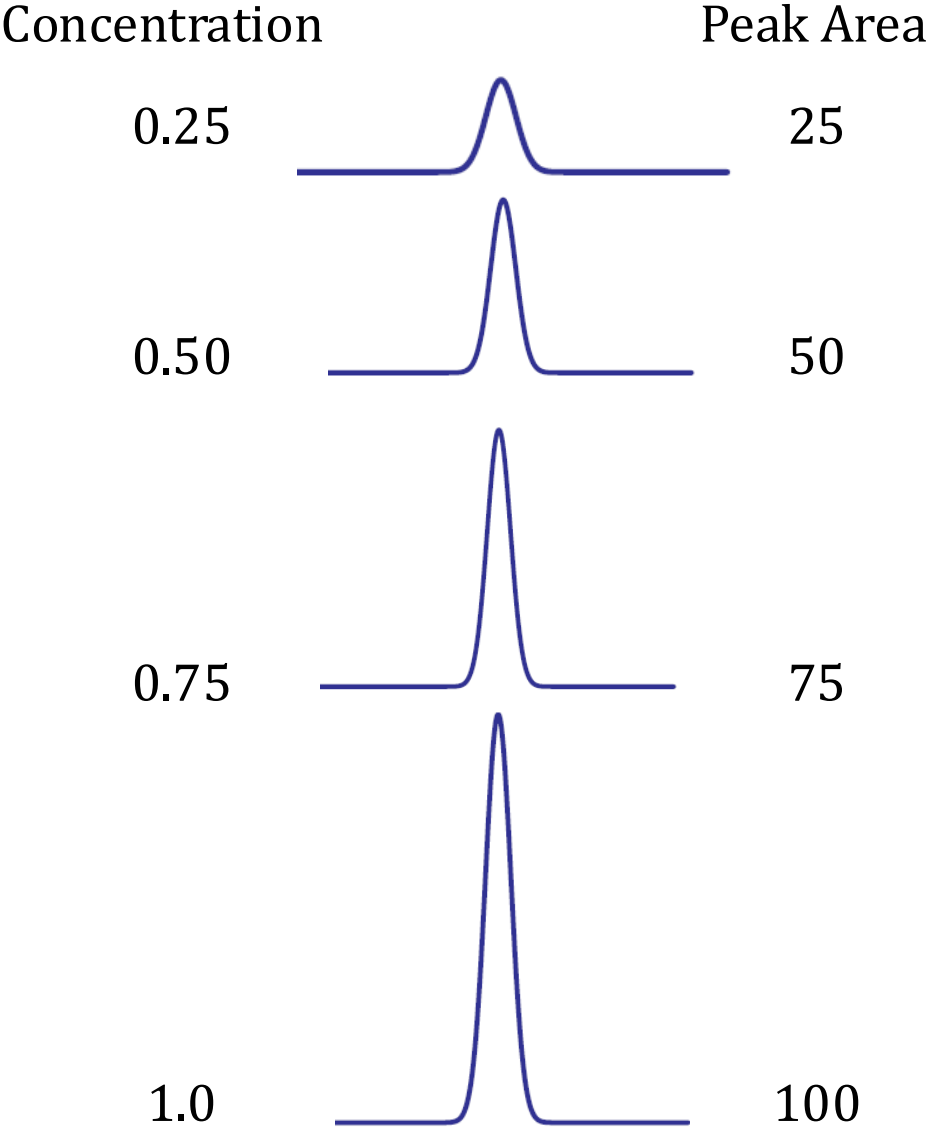
In the external standard method, we use one or more standards (ideally at least six independently prepared solutions with known analyte concentrations) to build **calibration curve**.

When the curve is linear, the slope (m) indicates constant sensitivity across the concentration range.

However, if the curve is nonlinear, the method's sensitivity varies with concentration, making the result more challenging to interpret.

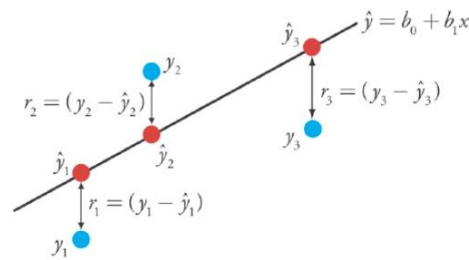
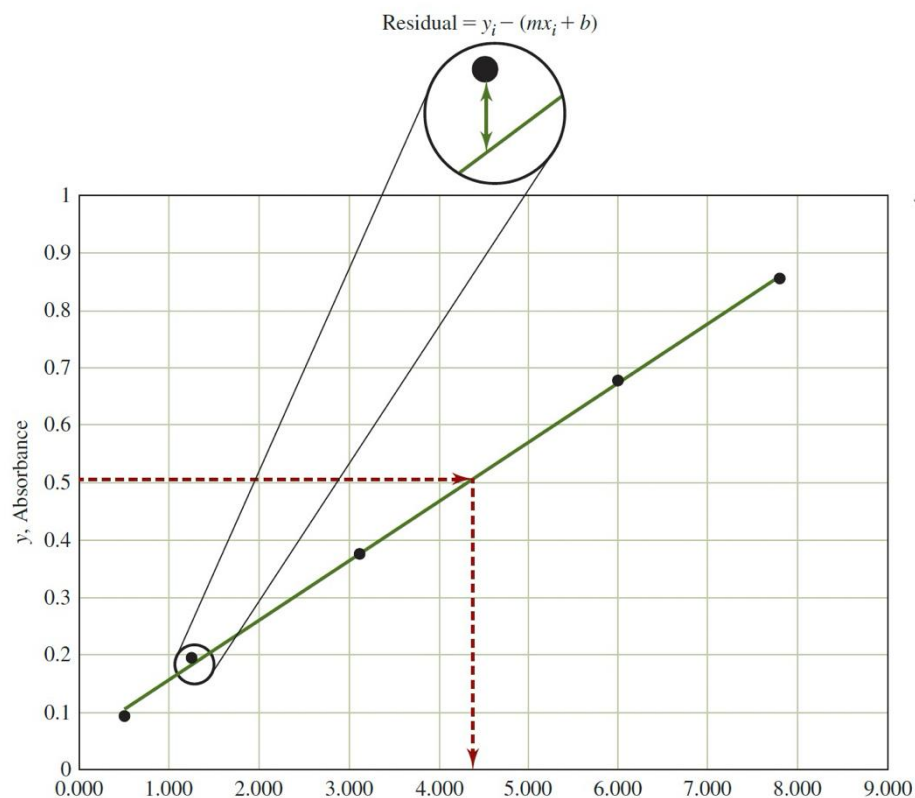


Calibration Curve



Linear Regression

Due to indeterminate errors, not all data points fall perfectly on the line. we use regression analysis to find the best-fit line through the data and for specifying the uncertainties.

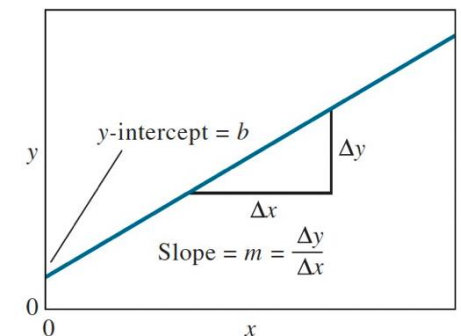


The Least-Squares Method

Assumption #1: There is a linear relationship between the measured response y and the standard analyte concentration x .

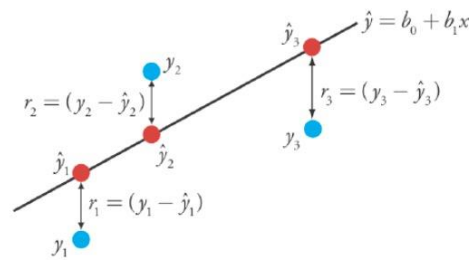
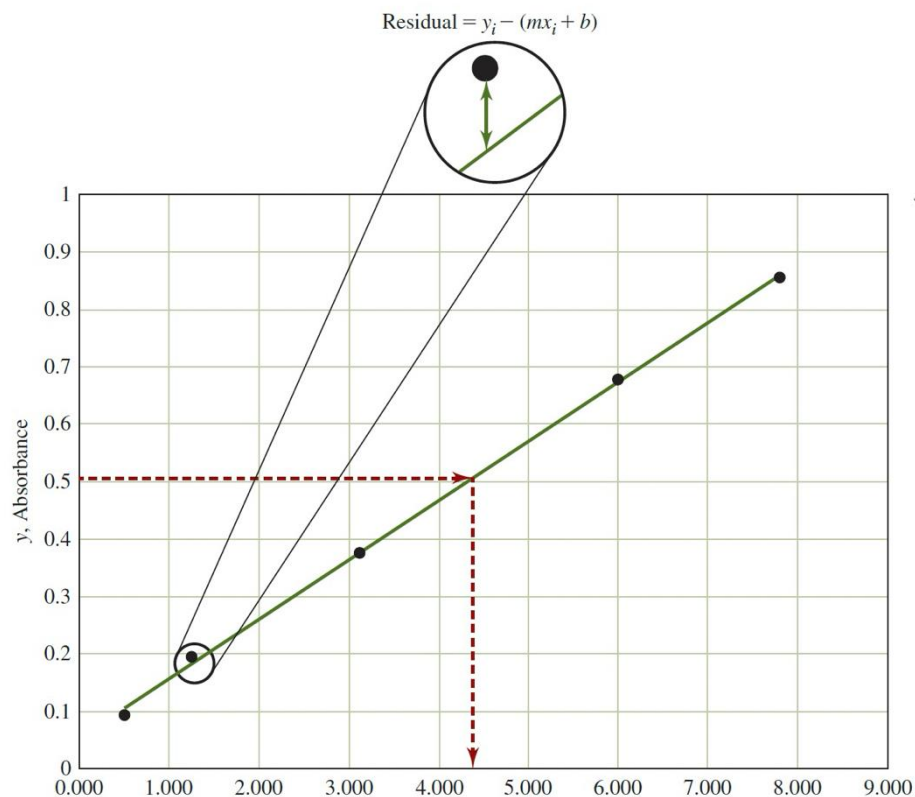
The regression model mathematically describes this assumption as:

$$y = mx + b$$



Linear Regression

Due to indeterminate errors, not all data points fall perfectly on the line. we use regression analysis to find the best-fit line through the data and for specifying the uncertainties.

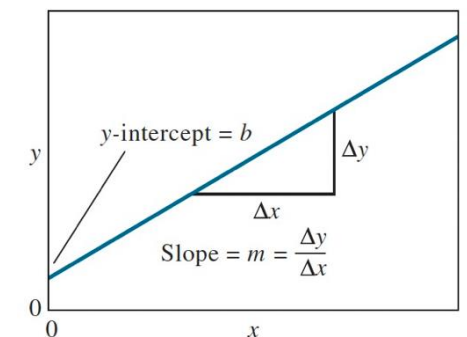


The Least-Squares Method

Assumption #2: any deviation of the individual points from the straight line results from error in the individual measurements.

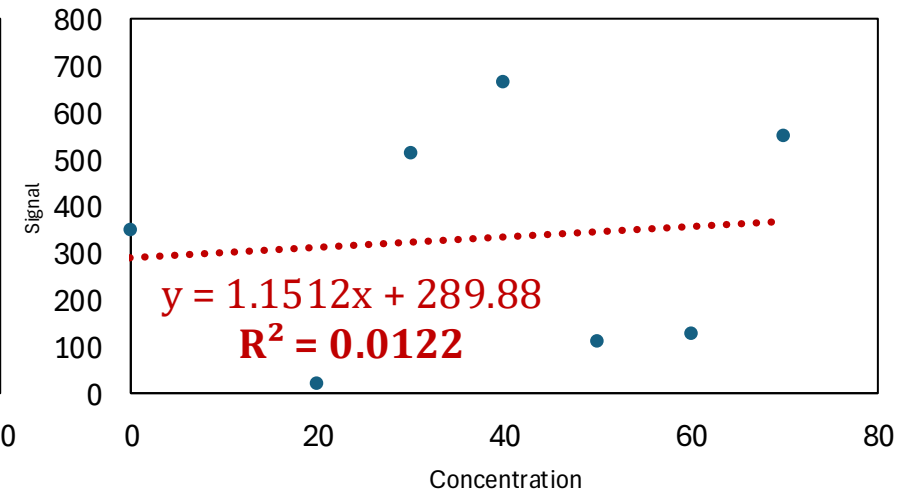
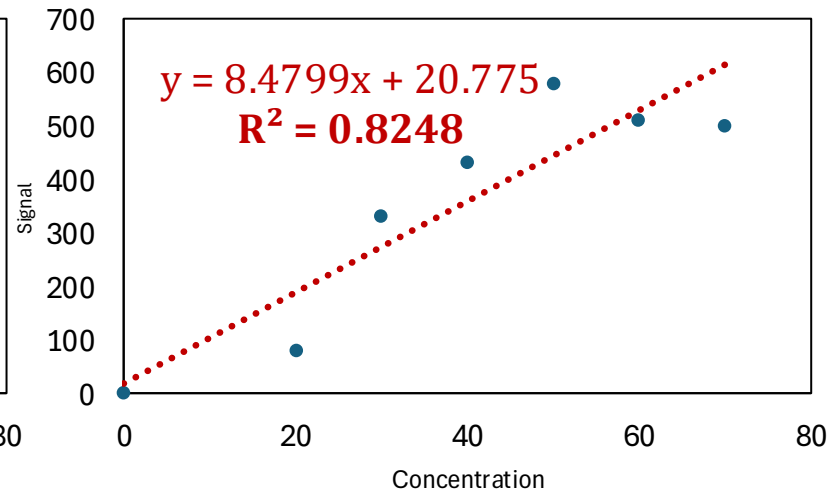
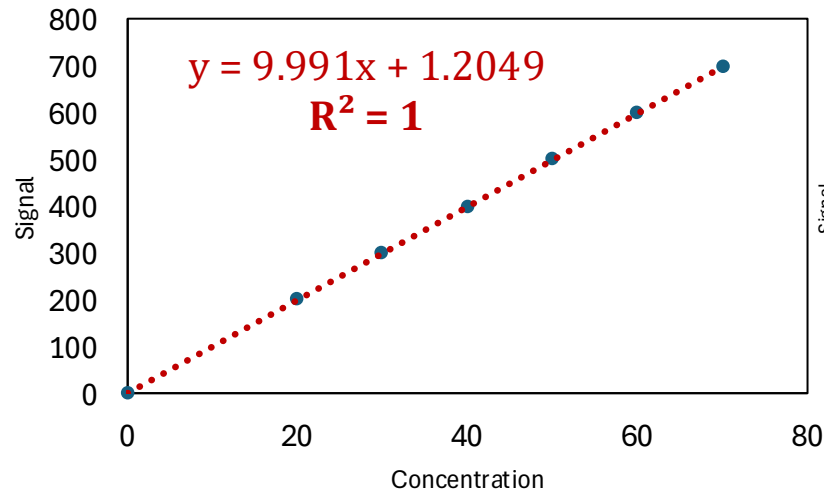
That is, assume that there is no error in x values of the points (concentrations).

$$y = mx + b$$



Least Squares

The goodness of fit of your best fit line is represented by R^2 . This is called the co-efficient of determination. The closer it is to 1, the better the value.

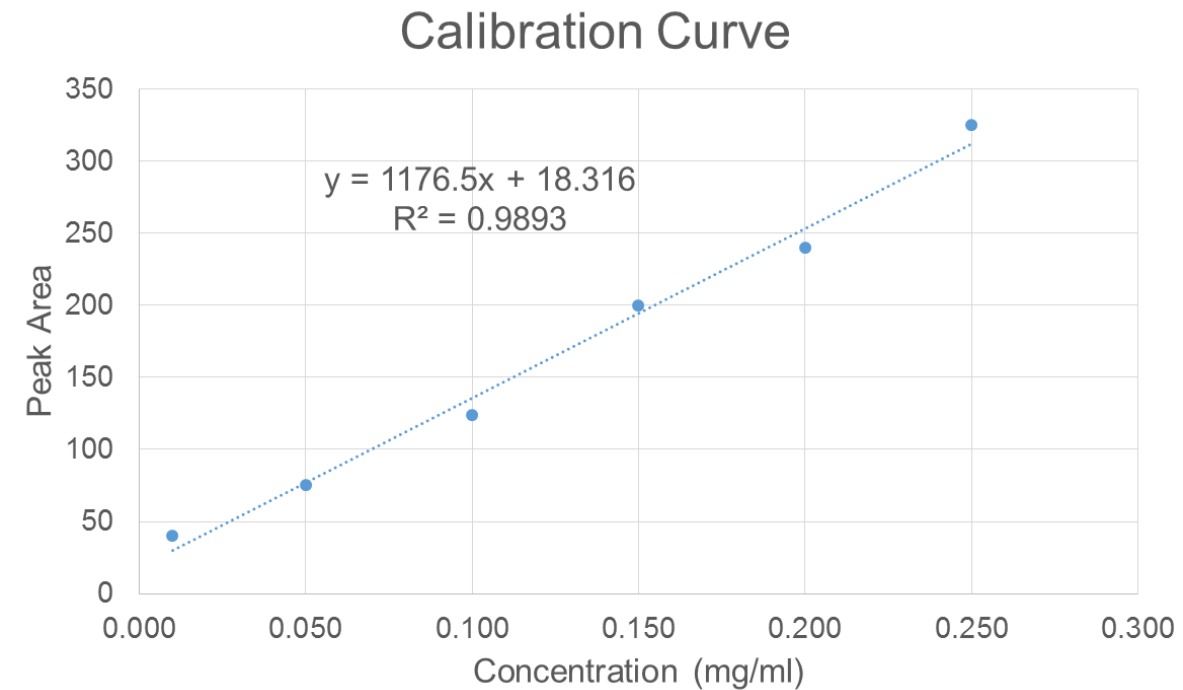


Exercises in Excel: [Link](#)

Practice Problem: Calibration Curve

Determine the concentration of benzene with a peak area of 184 using a calibration curve. The external standards gave the following responses:

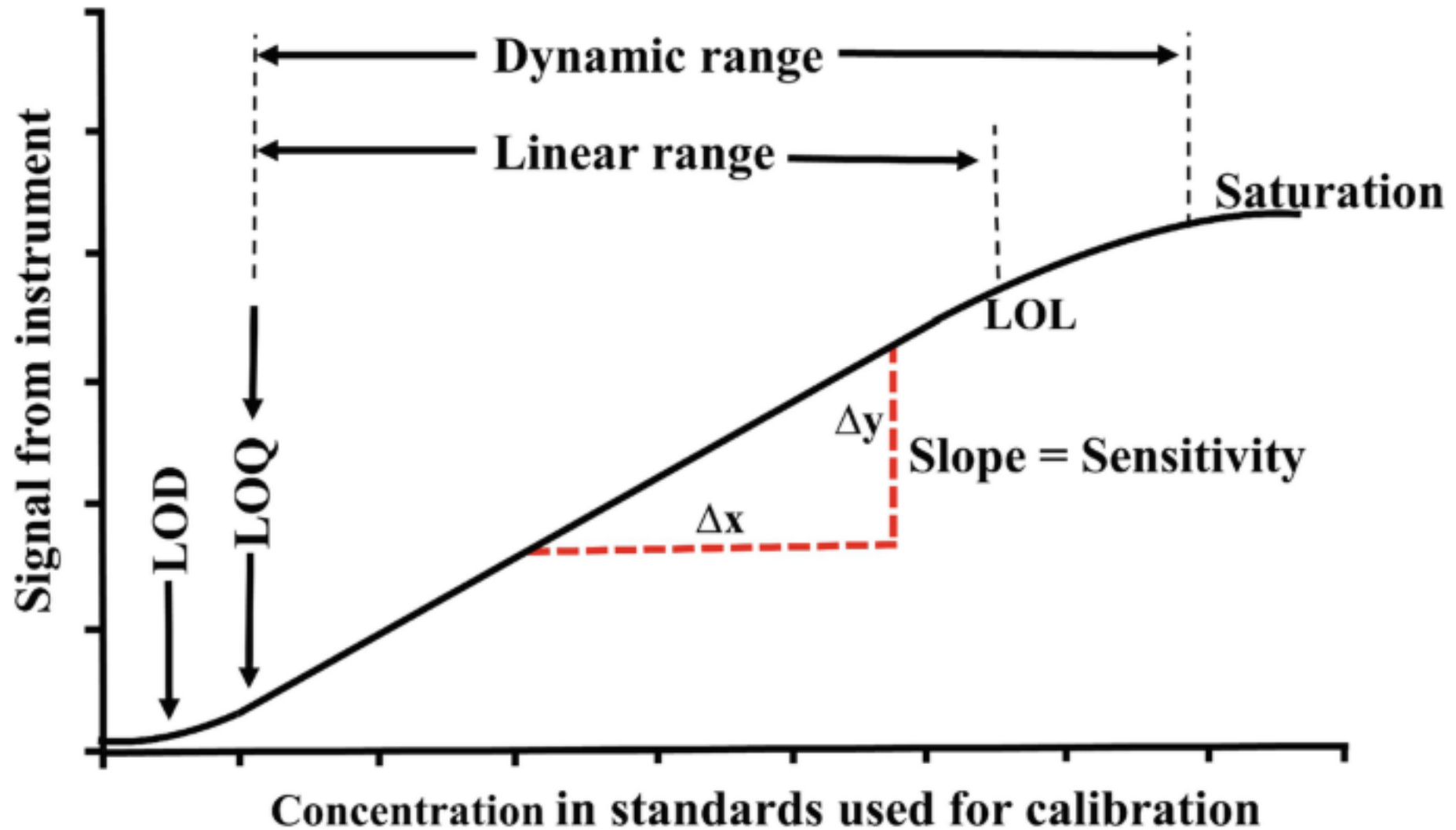
Concentration (mg/ml)	Peak Area
0.0100	40
0.0500	75
0.100	124
0.150	200
0.200	240
0.250	325



Practice Problem: Calibration Curve

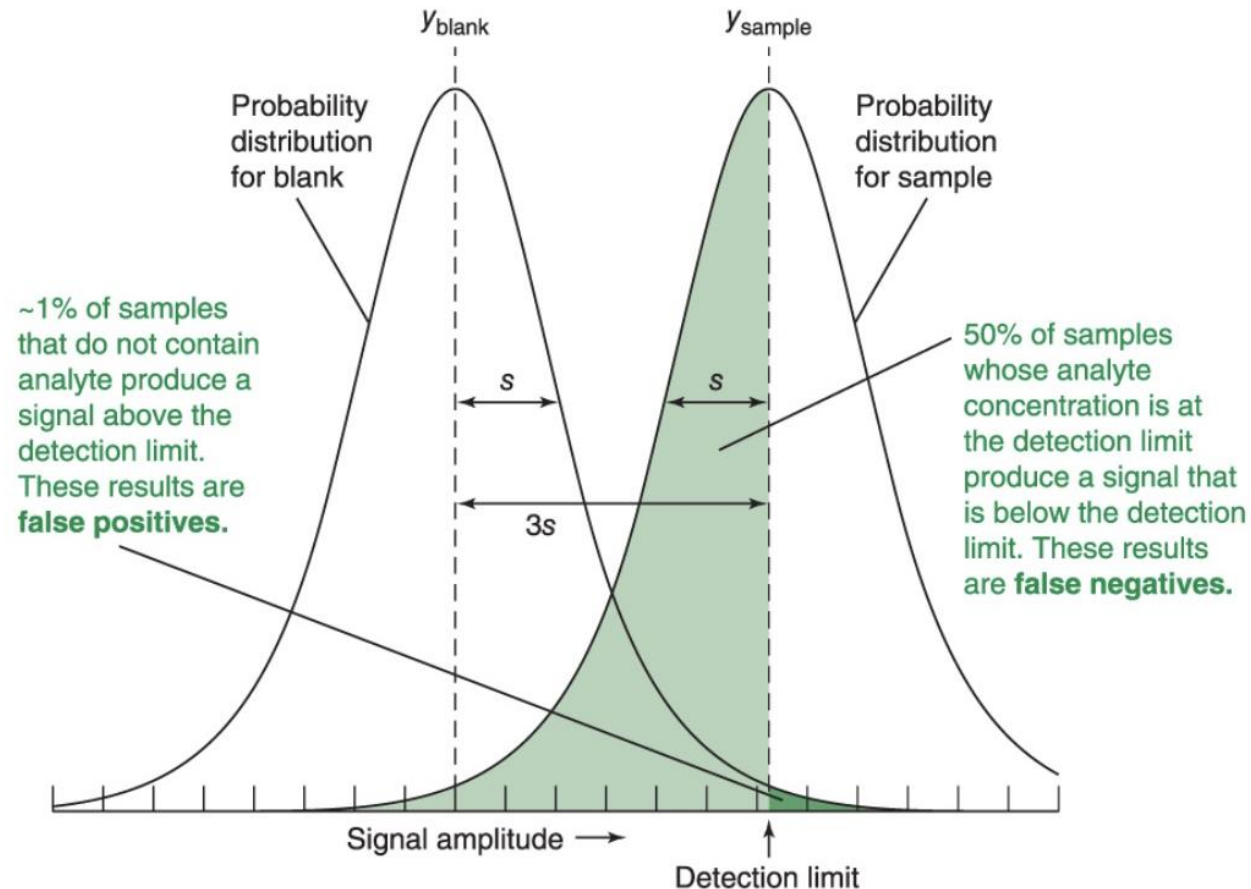


Figures of Merit for Analytical Methods



Limit of Detection

Limit of detection (LOD): the lowest concentration we can **report with a defined level of confidence**; meaning it's significantly different from a blank measurement.



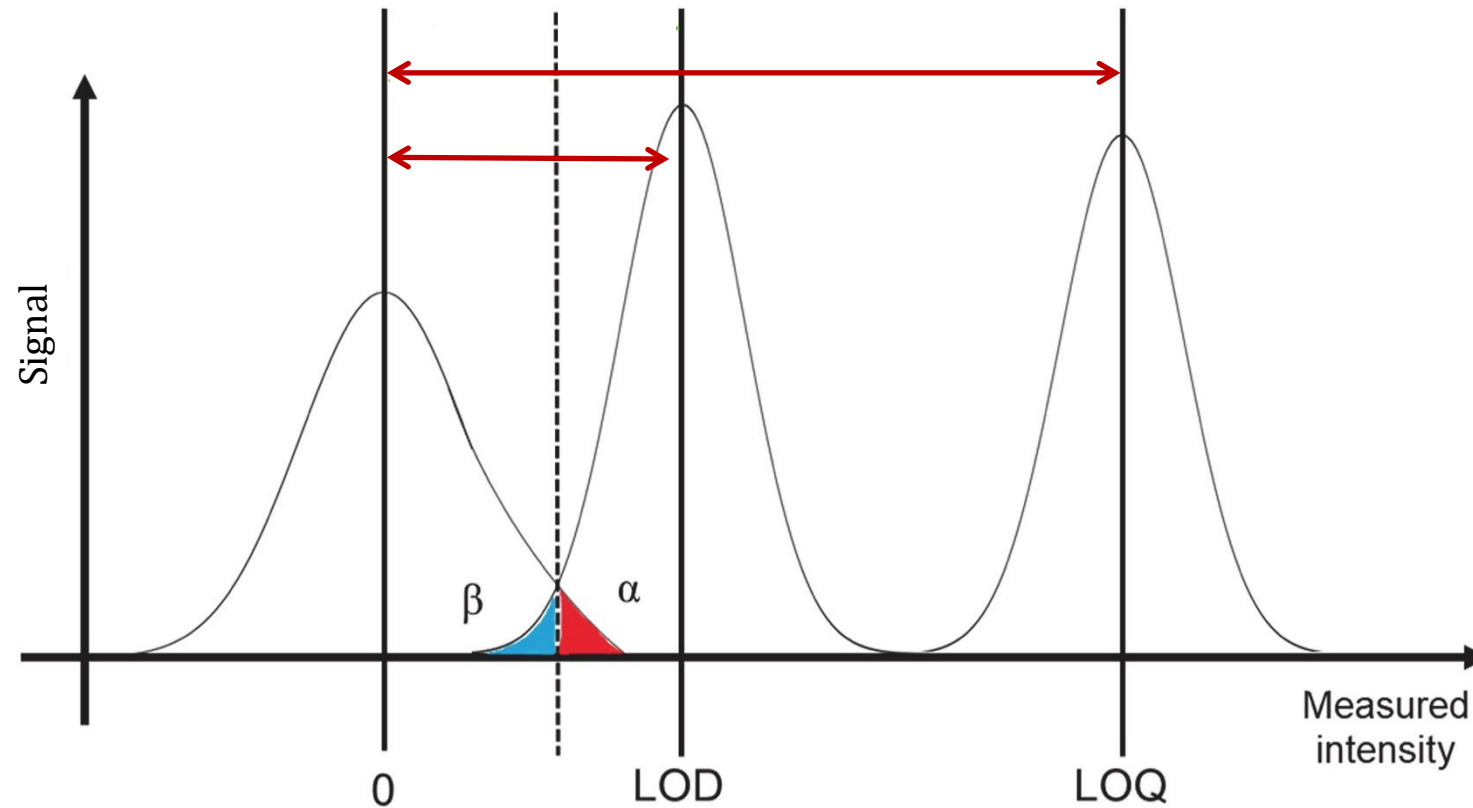
Harris/Lucy, *Quantitative Chemical Analysis*, 10e, © 2020 W. H. Freeman and Company

To establish the LOD with approximately 99% confidence, we define it so that only about 1% of truly blank samples produce a signal above the detection limit. In other words, we set a 1% **false positive rate**.

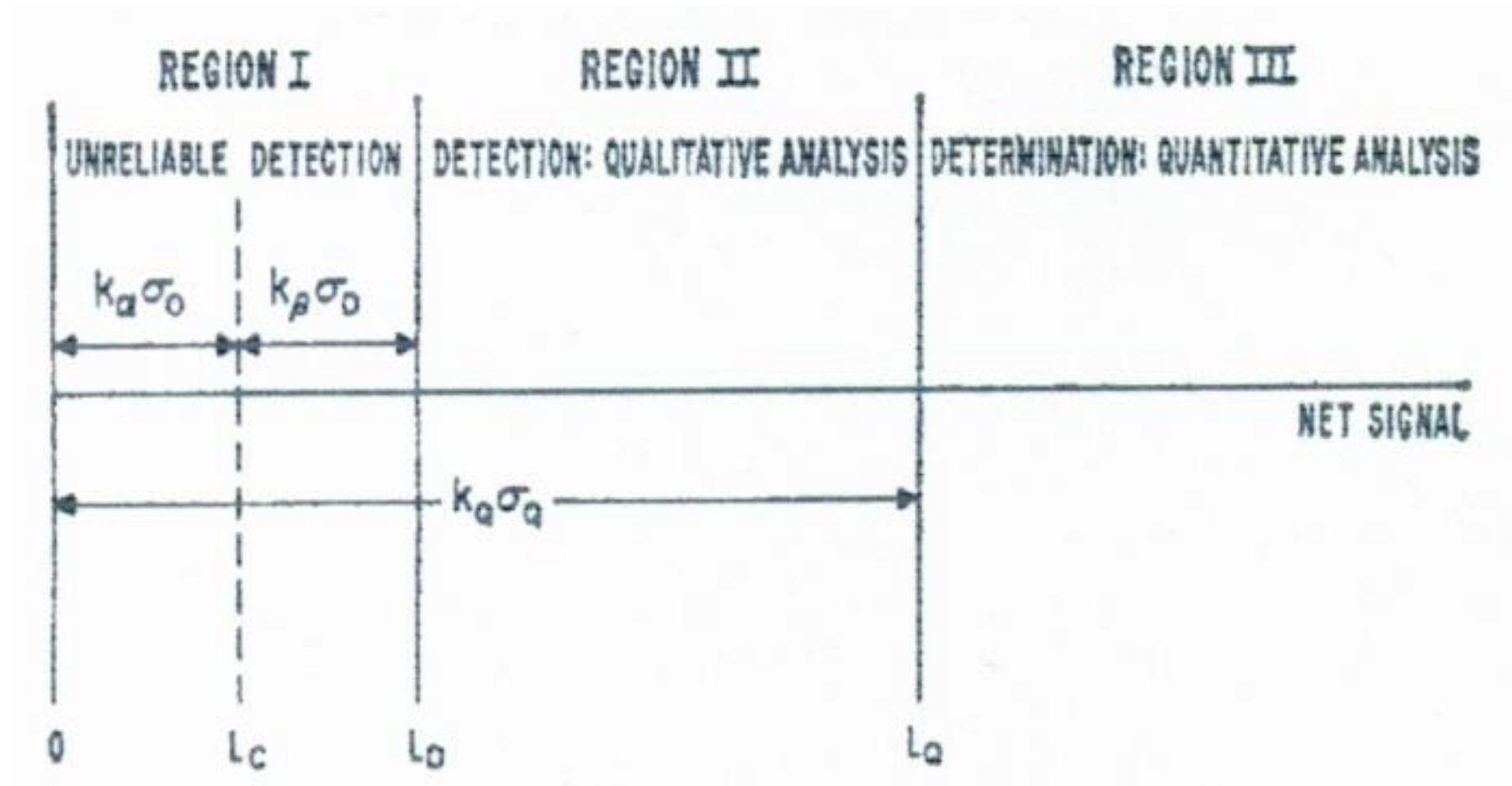
However, Note that, this detection limit gives us only 50% confidence in identifying a real signal from a sample that actually contains analyte at the LOD. That means half the samples at the detection limit will fail to give a detectable signal, leading to a **50% false negative rate** at that concentration.

Limit of Qualification

Limit of Qualification (LOQ): the smallest amount that can be measured with reasonable accuracy.



The Three Main Analytical Regions

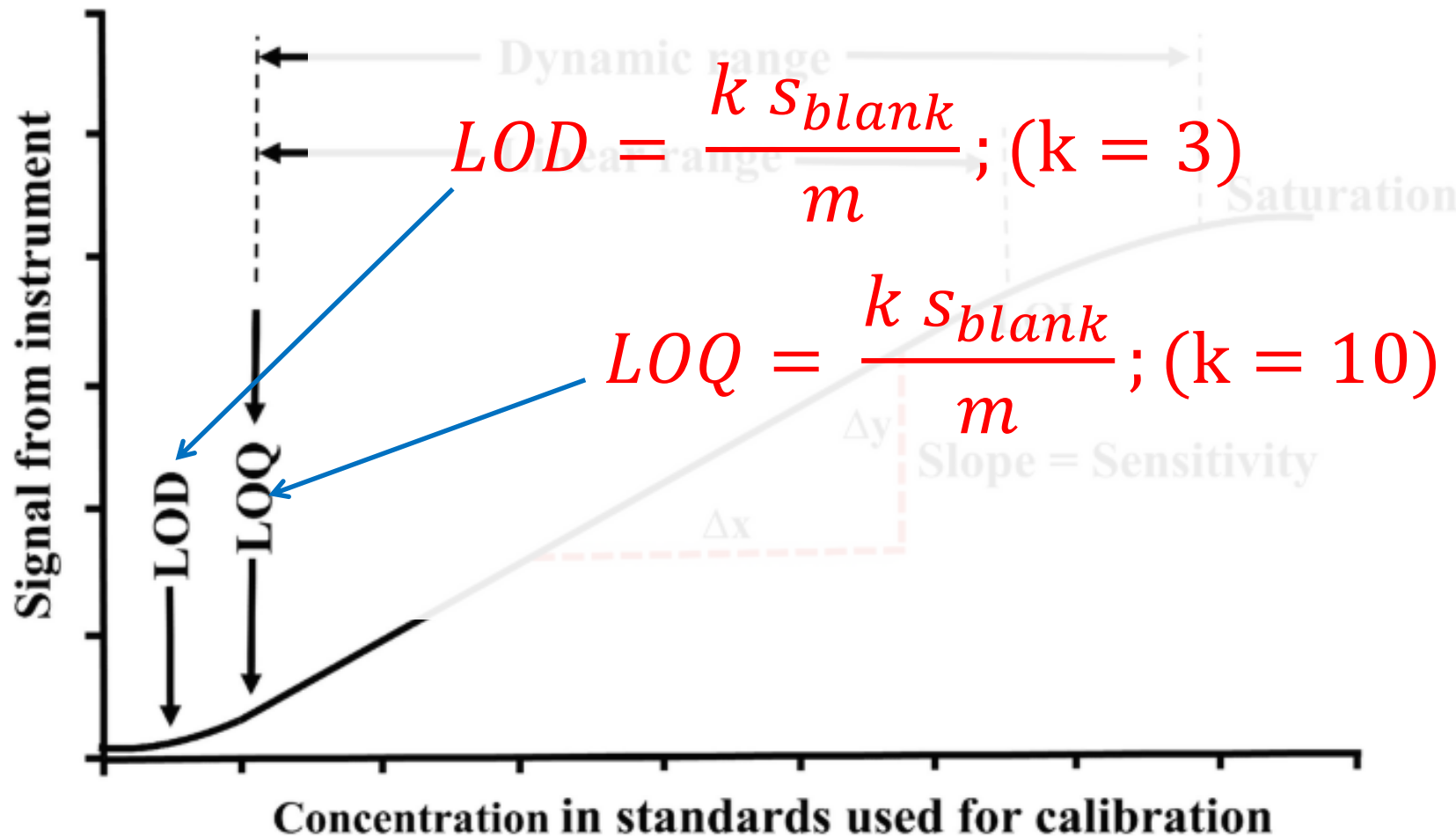


$k=1$ $k=3$

$k=10$

Limit of Detection

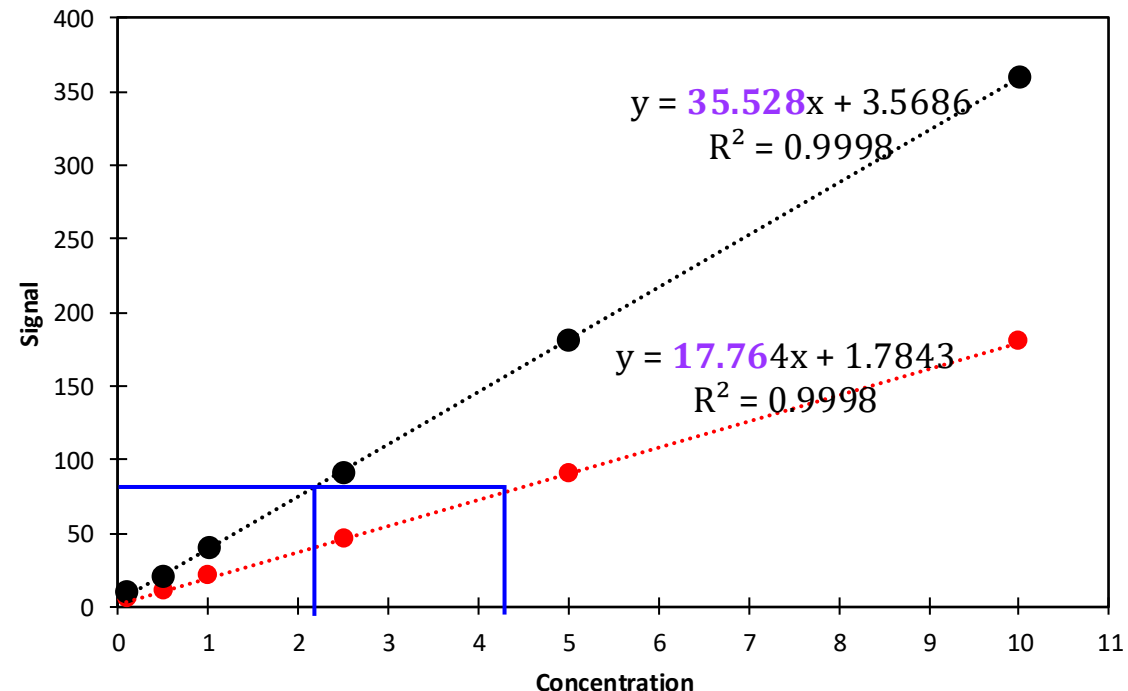
The limit of detection (LOD): the lowest concentration we can report with a defined level of confidence; meaning it's significantly different from a blank measurement.



Sensitivity

Calibration sensitivity is the slope of the calibration curve and remains constant when the curve is linear. However, it doesn't reflect how well we can distinguish between concentrations. For that, we use **analytical sensitivity**: the slope divided by the standard deviation of the signal at a given concentration. Unlike calibration sensitivity, analytical sensitivity often depends on concentration.

Concentration	Signal 1	Signal 2
0.1	5	10
0.5	10	20
1	20	40
2.5	45	90
5	90	180
10	180	360

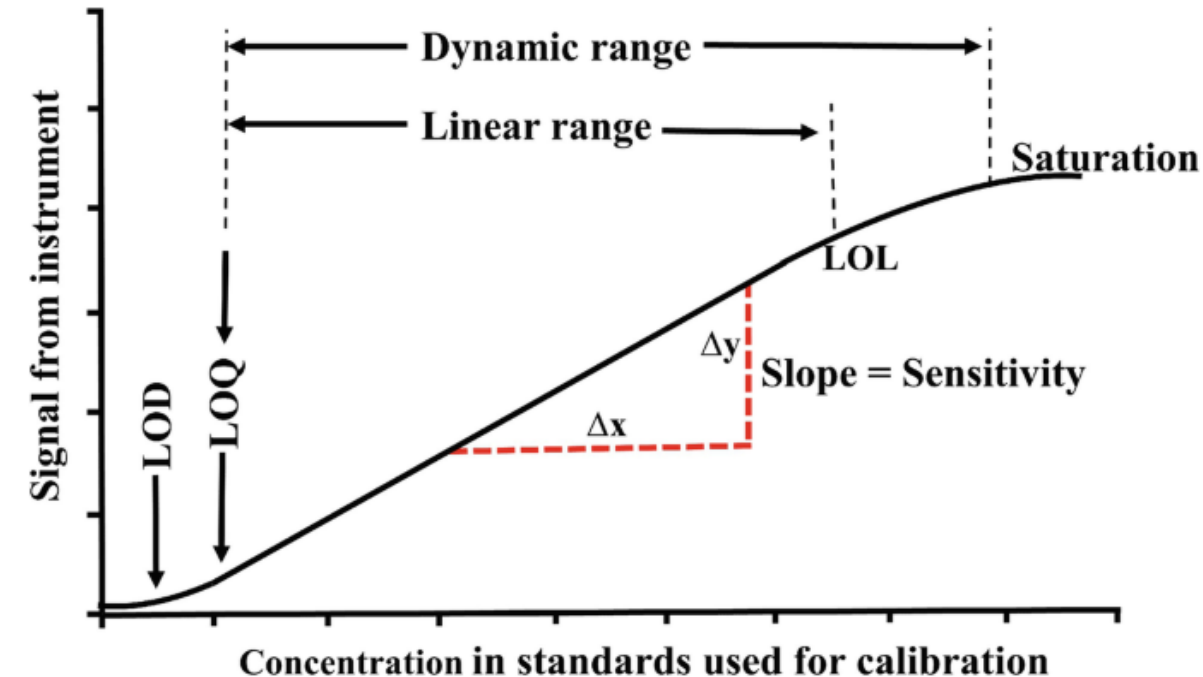


Dynamic Range vs. Linear Dynamic Range

Dynamic Range

The full range of analyte concentrations **where the instrument gives a measurable response**—not necessarily linear.

It extends from the LOD (or LOQ) to the point where the signal plateaus or becomes unreliable (e.g., due to saturation).



Linear Range

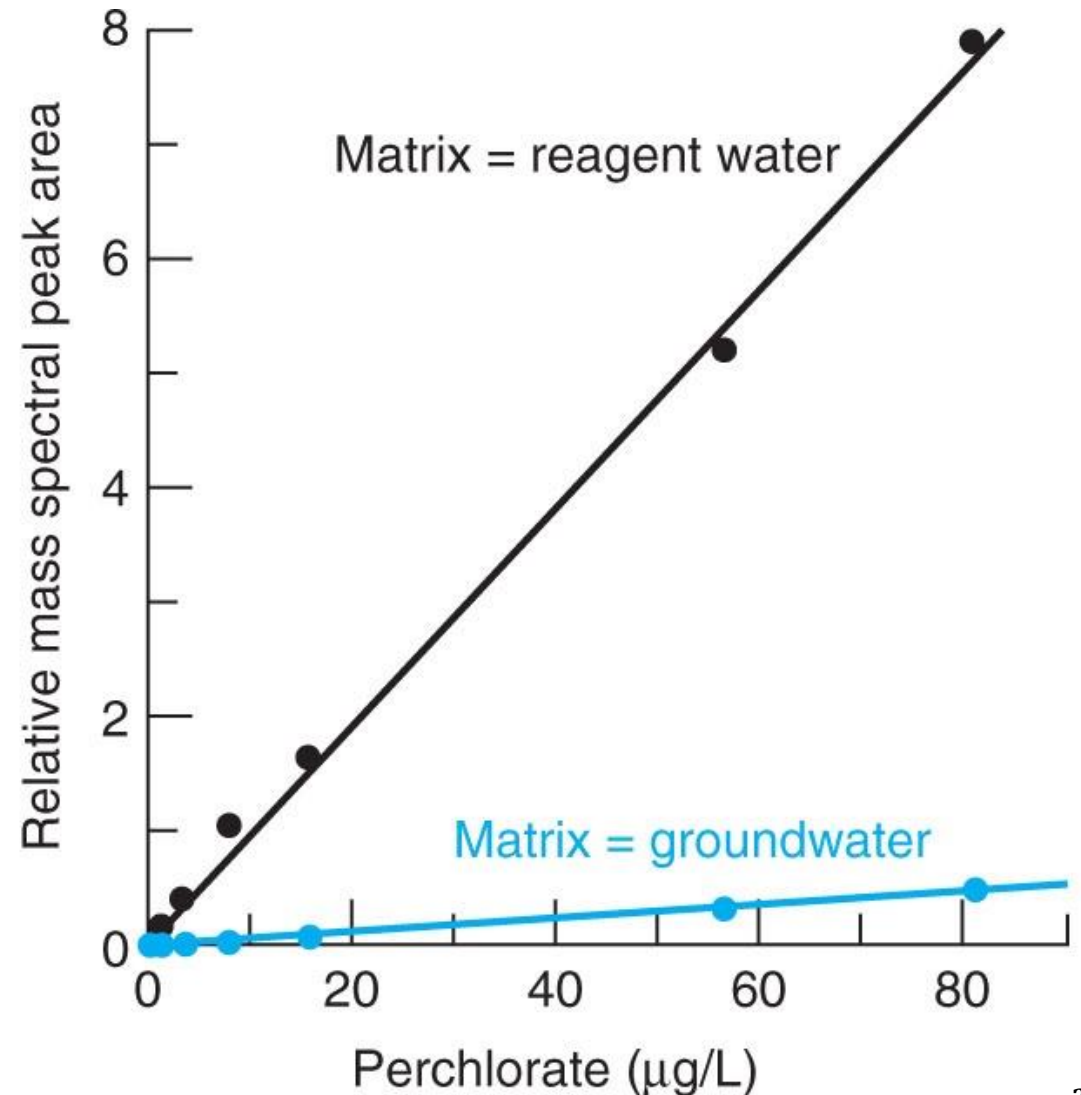
The portion of the dynamic range where the response is directly proportional to concentration. The lower limit **LOD** (or LOQ).

The upper limit is where the signal or slope deviates from linearity (**LOL**)—usually by more than 5%.

Deviations at high concentrations often result from nonideal detector behavior or chemical effects.

External Standard Addition Method (ExSTA)

Helps **correct for matrix** effects: ideal when we know what analyte we're targeting, but the sample composition is complex, variable, or unknown, make it difficult or impossible to prepare standards or blanks that match the **sample matrix**: interferences from other components that affect the analytical signal or slope of the calibration curve.



ExSTA: Single-Point Method

The signal of the unknown sample is measured first. A known amount of standard is then added to the same solution and the signal is measured again. Because analytical signal (I) is proportional to concentration (C): $I \propto C$ (or $I = kC$)

Signal from the original sample:

$$I_X = k[X]_i$$

Signal after adding standard:

$$I_{S+X} = k([S]_f + [X]_f)$$

Taking the ratio of the two signals:

$$\frac{[X]_i}{[S]_f + [X]_f} = \frac{I_X}{I_{S+X}}$$

In this method we **need to account for dilutions** for $[S]_f$ and $[X]_f$:

$$[S]_f = [S]_i \left(\frac{V_s}{V} \right)$$

$$[X]_f = [X]_i \left(\frac{V_x}{V} \right)$$

$[X]_i$ = initial concentration of sample

$[S]_f$ = final concentration of standard

$[X]_f$ = final concentration of sample

I_X = intensity/signal from sample

I_{S+X} = intensity/signal from sample and standard

V = total volume after addition

ExSTA: Single-Point Method

The signal of the unknown sample is measured first. A known amount of standard is then added to the same solution and the signal is measured again. Because analytical signal (A) is proportional to concentration (C_u): $A \propto C_u$ (or $A = kC_u$)

Signal from the original sample: $A_1 = kC_u$

Signal after adding standard: $A_2 = k(C_uV_u + C_sV_s)/V_t$

Solving for the unknown concentration:
$$C_u = \frac{A_1C_sV_s}{A_2V_t - A_1V_u}$$

In this form, the **volume correction is already included** in the equation, so dilution does not need to be handled separately.

C_u : concentration of analyte in the unknown

C_s : concentration of standard

A_1 & A_2 (or S): measured signals

K : proportionality constant

V_s : volume of standard added

V_u : volume of unknown

V_t : total volume after addition ($V_u + V_s$)

Practice Problem

Solution believed to contain PCP gives a UV-Vis absorbance reading of 0.35. 20 mL of a standard of PCP, with a concentration of 0.5 mg/mL was added to 80 mL of sample. A new reading was taken and gave a reading of 0.92. What was the initial concentration of PCP?

$I_x = 0.35$ (initial absorbance)

$I_{s+x} = 0.92$ (absorbance after spike)

$V_x = 80$ mL (sample volume)

$V_s = 20$ mL (standard added)

$V = V_x + V_s = 100$ mL (final total volume)

$[S]_i = 0.5$ mg/mL



Practice Problem

Solution believed to contain PCP gives a UV-Vis absorbance reading of 0.35. 20 mL of a standard of PCP, with a concentration of 0.5 mg/mL was added to 80 mL of sample. A new reading was taken and gave a reading of 0.92. What was the initial concentration of PCP?

$A_1 = 0.35$ (initial absorbance)

$A_2 = 0.92$ (absorbance after spike)

$V_u = 80$ mL (sample volume)

$V_s = 20$ mL (standard added)

$V_t = V_x + V_s = 100$ mL (final total volume)

$C_s = 0.5$ mg/mL



Practice Problem

A sample solution of unknown Ni^{2+} gives an UV-Vis absorbance of $A_1 = 0.246$.

After adding 2.00 mL of 20.0 ppm Ni^{2+} standard to 10.00 mL of the sample and diluting to $V_t = 12.00$ mL, the signal rises to $A_2 = 0.412$.

Find the unknown Ni^{2+} concentration in the original sample.

ExSTA: Multiple-Point Method

Instead of adding standard once, **several additions of standard** are made and the signal is measured after each addition.

Advantages:

- improves precision and accuracy
- allows graphical determination of the unknown concentration
- provides a check for linearity of the analytical response

Two common experimental implementations:

1. Successive additions to a single solution
2. Multiple solutions with constant final volume

ExSTA: Multiple-Point Method

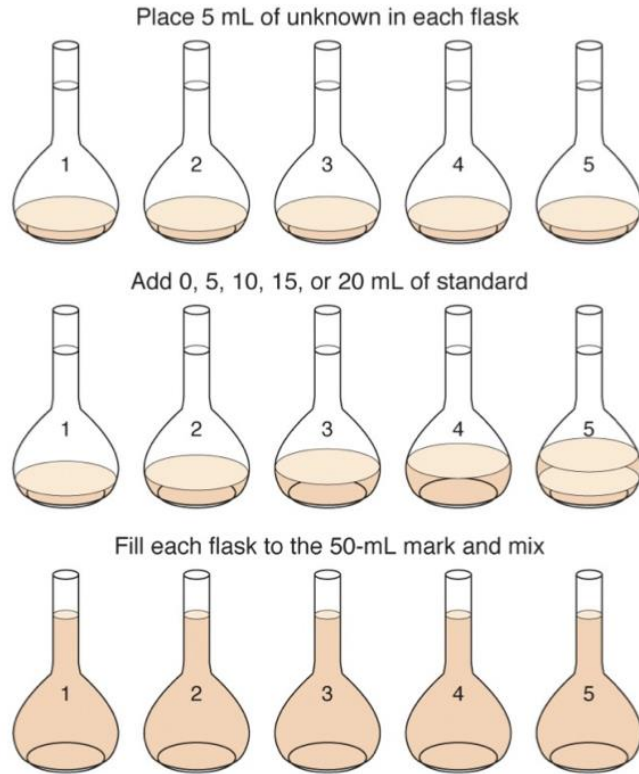
Feature	Multiple Solutions (Constant Final Volume)	Successive Additions to One Solution
Setup	Separate flasks prepared with different standard additions	Standard added repeatedly to the same sample solution
Number of solutions	Several independent solutions	One solution measured repeatedly
Total volume	Constant final volume	Changes after each addition
Dilution effect	Already controlled	Must be corrected for
Data analysis	Linear regression calibration curve	Graphical method accounting for dilution
Typical use	Most common implementation in analytical chemistry	Instrument methods where repeated measurement is easy

Multiple Solutions with Constant Final Volume

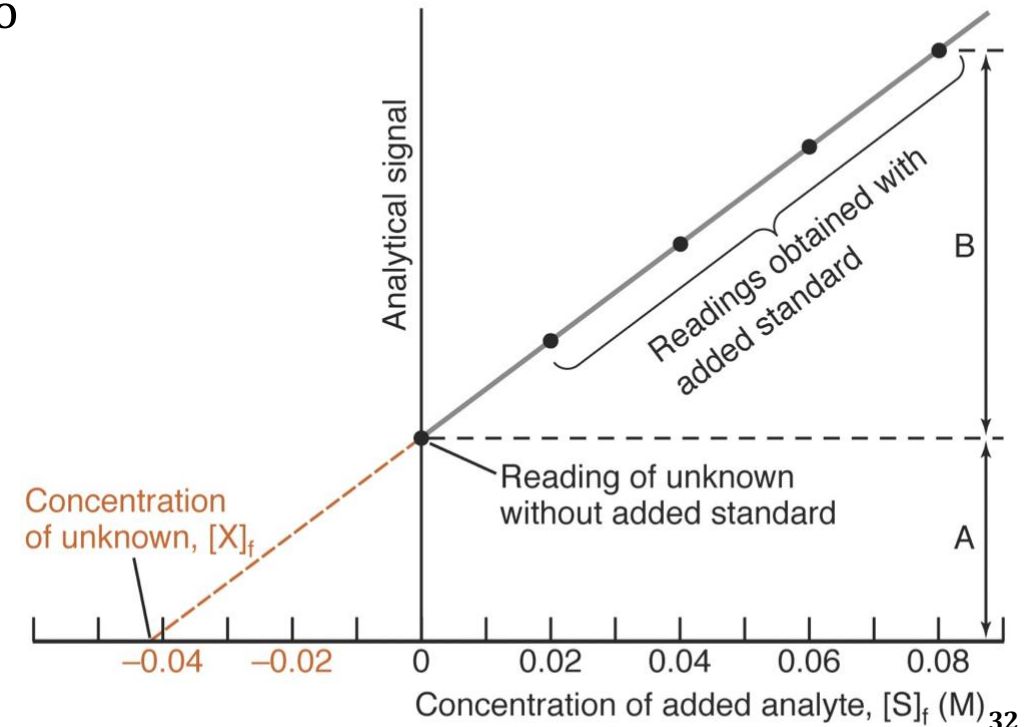
In this method, several separate solutions are prepared, each containing the **same volume of sample and increasing amounts of standard**. Each solution is then diluted to the **same final volume** before measuring the signal.

Procedure:

1. Pipette the same volume of unknown sample into several volumetric flasks.
2. Add different volumes of standard solution to each flask.
3. Dilute each flask to the same final volume.
4. Measure the signal for each so



Because every solution is diluted to the same final volume, the dilution factor is constant and the analytical signal varies linearly with the amount of added standard.



Multiple Solutions with Constant Final Volume

$$S = k \left(C_u \frac{V_u}{V_f} + C_s \frac{V_s}{V_f} \right) \xrightarrow[\text{equation gives a linear form}]{\text{Rearranging the signal}} S = \frac{kC_s}{V_f} V_s + \frac{kC_u V_u}{V_f}$$

which matches $\rightarrow y = mx + b$

From the slope and intercept of the calibration line

$$C_x = \frac{bC_s}{mV_u}$$

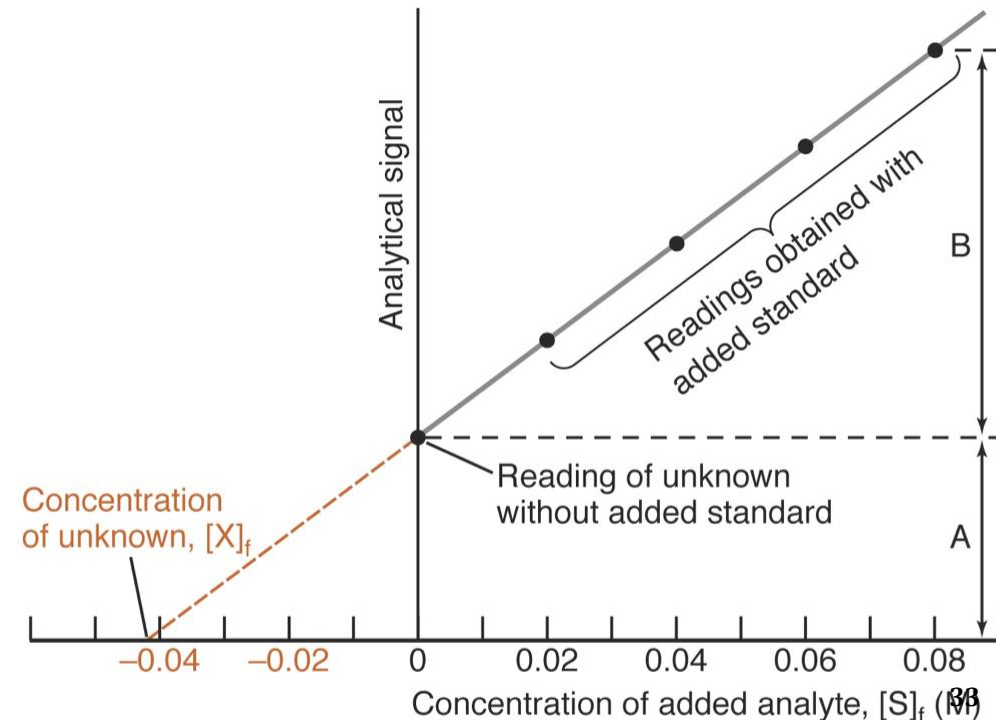
The x-intercept of the calibration curve corresponds to

$$X_{int} = -\frac{C_u V_u}{C_s}$$

V_u : the same volume of unknown sample

V_f : the same final volume in each flask

V_s : increasing volumes of standard



Practice Problem

A student analyzes an environmental water sample containing Cu^{2+} using UV-Vis spectroscopy at 620 nm. To correct for matrix effects each 25.00 mL aliquot of the sample is transferred to a 50.00 mL volumetric flask, and standard Cu^{2+} solution (10.0 ppm) is added in increasing amounts before dilution to the mark. Find the unknown concentration of Cu^{2+} in the sample (C_u).

Volume of standard added (mL)	Absorbance
0.00	0.215
2.00	0.339
4.00	0.461
6.00	0.583
8.00	0.709

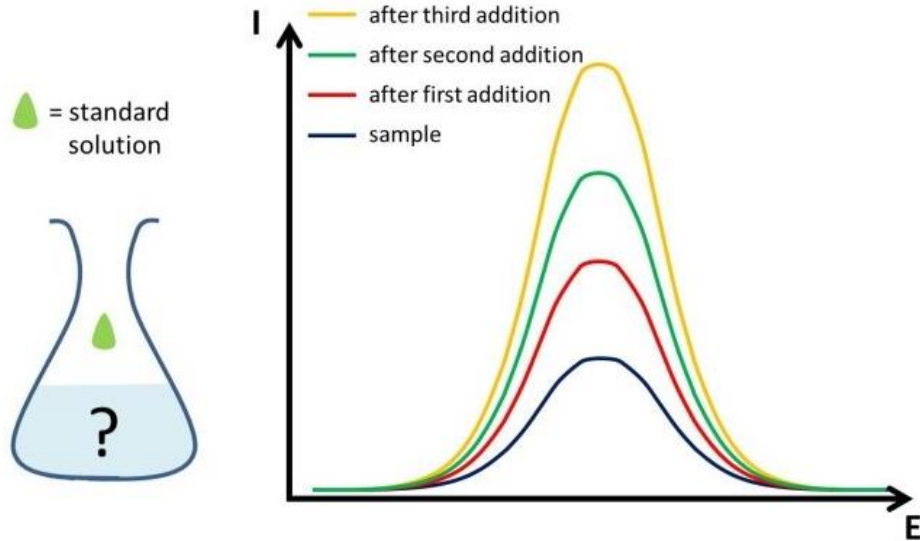
Practice Problem

0.8

0.7

9

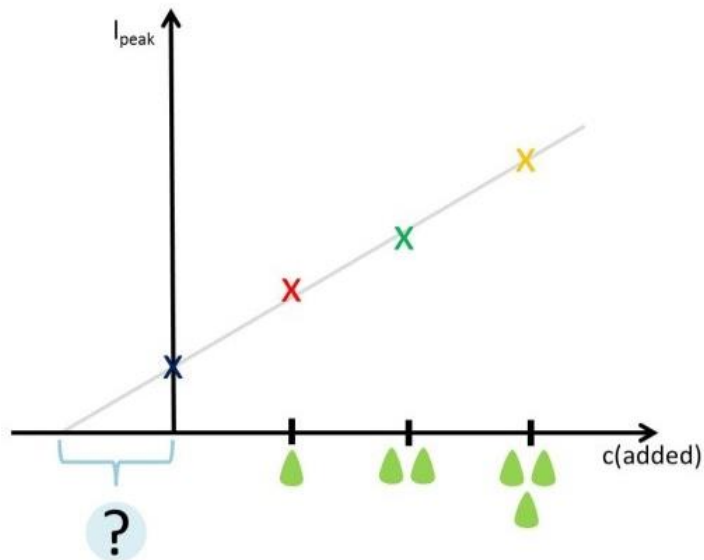
Successive Additions to One Solution



In this method, the standard solution is added successively **to the same sample solution**.

Procedure:

1. Measure the signal of the original sample.
2. Add a small volume of standard.
3. Measure the signal again.
4. Repeat several additions.



Successive Additions to One Solution

$$S = k \left(\frac{C_u V_u + C_s V_s}{V_u + V_s} \right)$$

For the original unspiked sample: when $V_s = 0 \rightarrow S_0 = kC_u$

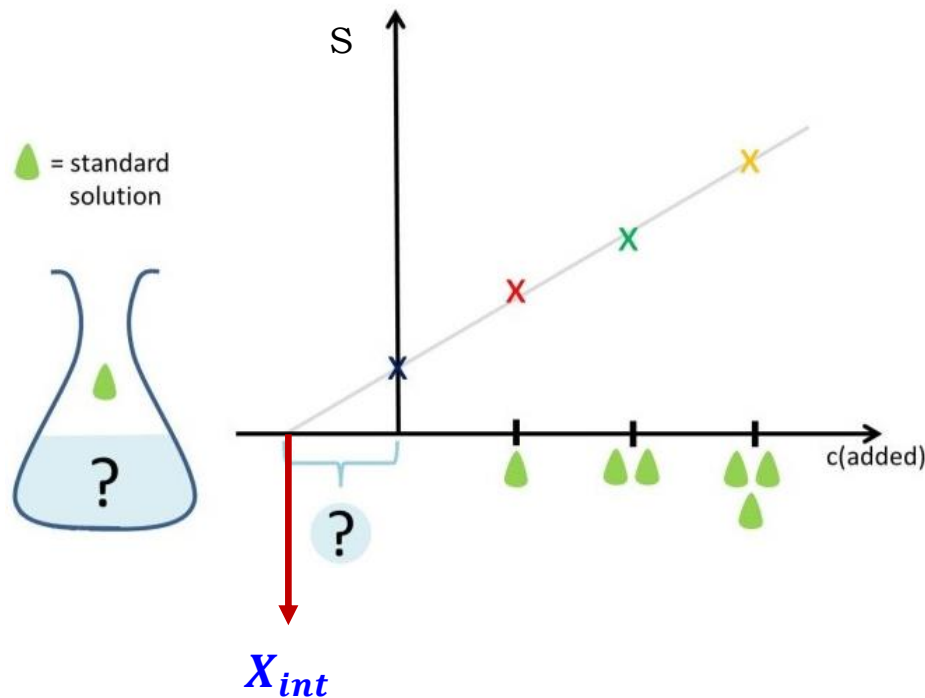
By extrapolating the calibration line to the x-axis, we determine the negative of the analyte concentration in the original sample.

Meaning of the x-intercept: The x-intercept (X_{int}) represents the amount of added standard required to reduce the signal to zero.

If x-axis = added standard concentration: $C_u = -X_{int}$

If x-axis = added standard volume:

$$C_u = \frac{-X_{int} C_s}{V_u}$$

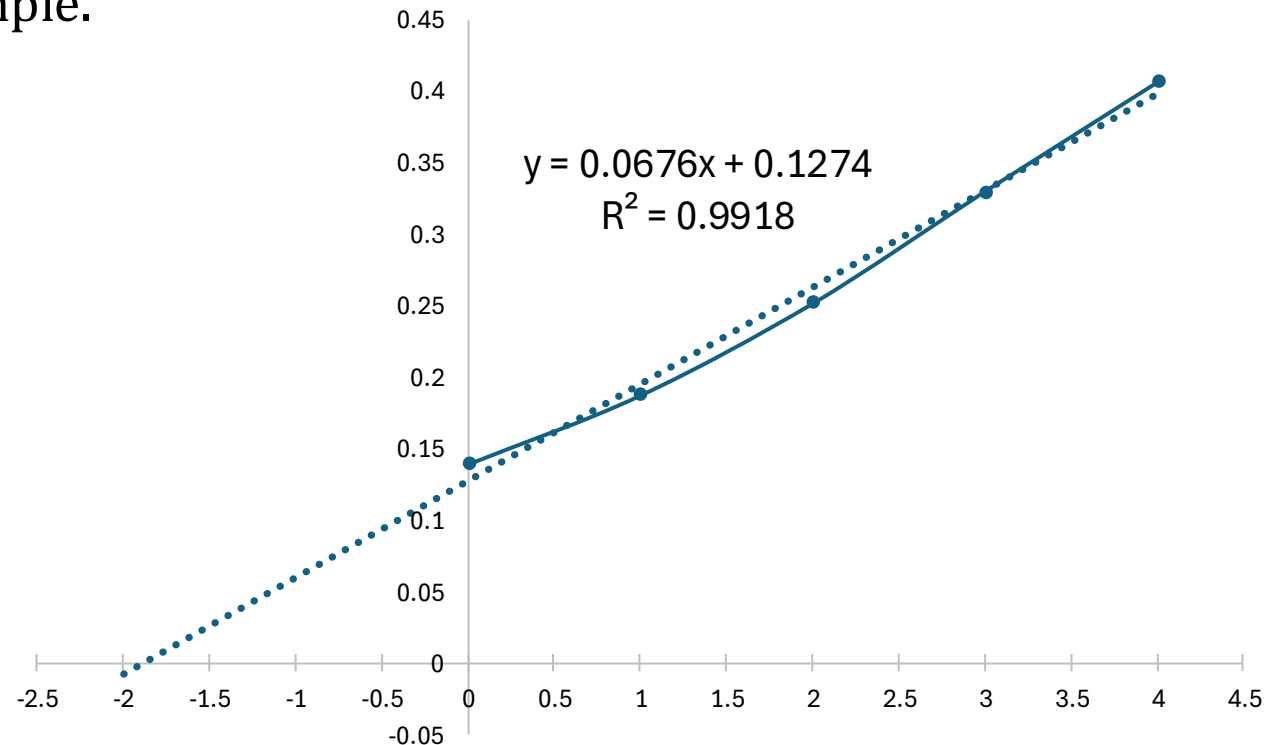


Practice Problem

A researcher is tasked with determining the concentration of oxybenzone in a sunscreen sample using UV-Vis spectroscopy. Because the sunscreen matrix contains oils, emulsifiers, and other UV-absorbing compounds, a significant matrix effect is suspected.

To account for this, increments of a 10.0 ppm oxybenzone standard solution are added directly to 25.00 mL of the unknown sample and the absorbance is measured after each addition. The following data were obtained. Calculate the concentration of oxybenzone in the original sample.

Volume of standard added (mL)	Absorbance
0.00	0.220
1.00	0.287
2.00	0.352
3.00	0.419
4.00	0.486



Practice Problem



Standard Addition Disadvantages

Math is a bit more complicated

Does not account for random errors

Variations in instrument response from run-to-run

Variations in quantity of sample analyzed from run-to-run (e.g., GC- MS)

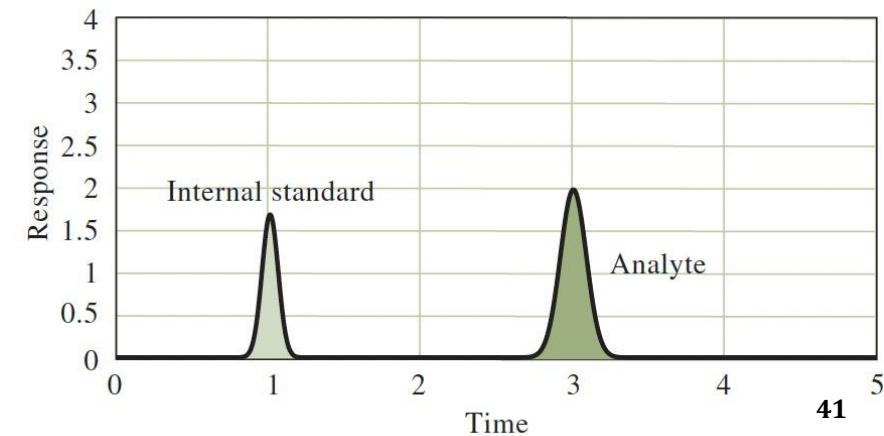
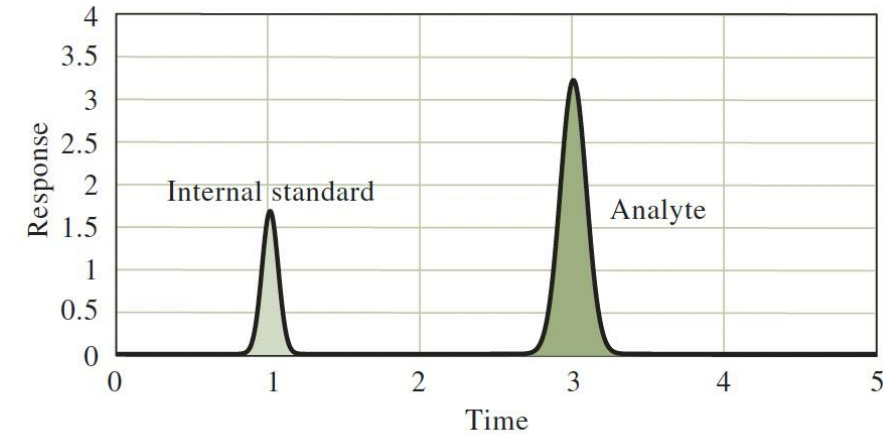
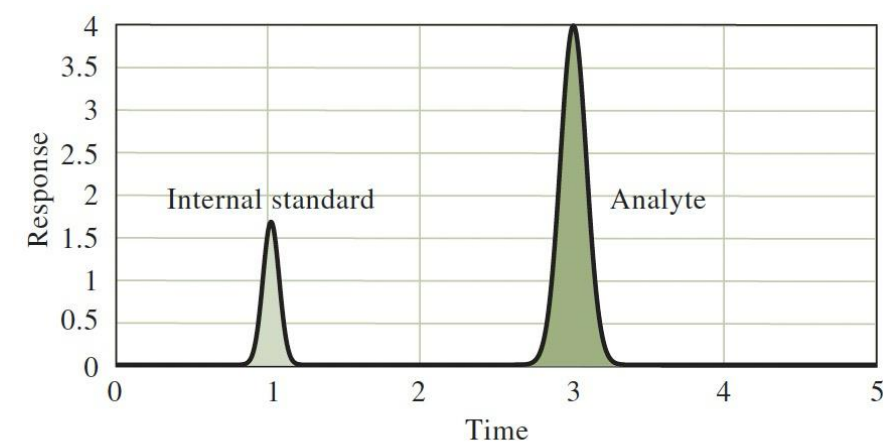
Internal Standard

Provides an independent signal which can be used for normalization

Known concentration of Internal standard added to each sample and calibration sample

Ratio of standard response to analyte helps to overcome discrepancies between injections (random error)

Curve of ratios of peak area (sample:IS) plotted vs [Sample]:[IS]



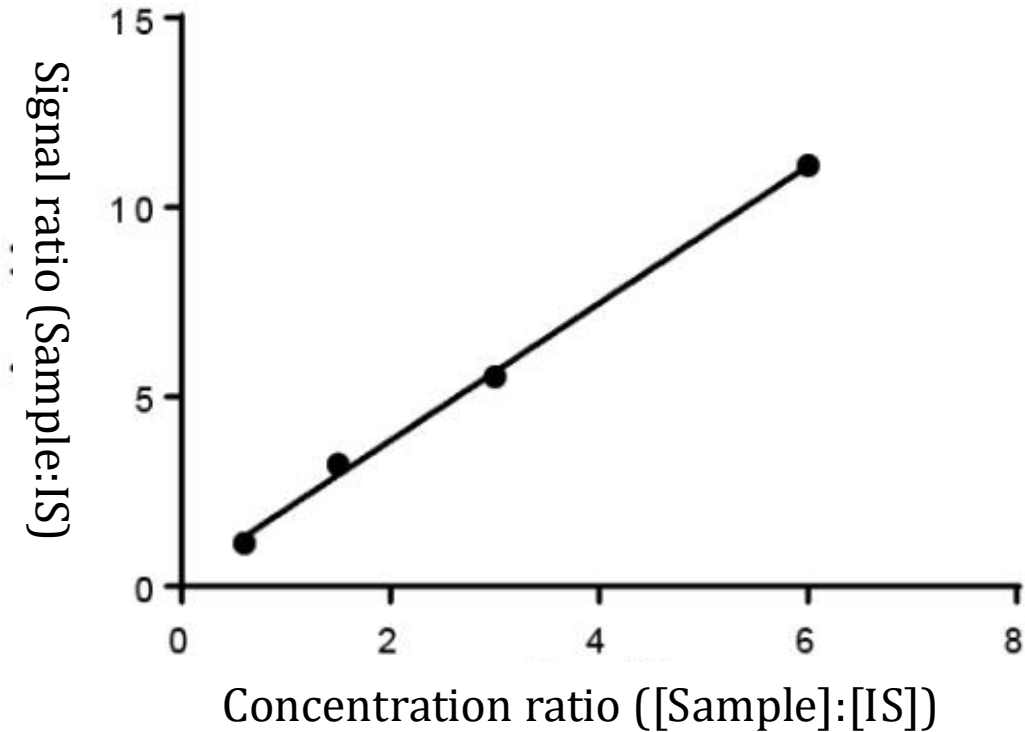
Internal Standard

Provides an independent signal which can be used for normalization

Known concentration of Internal standard added to each sample and calibration sample

Ratio of standard response to analyte helps to overcome discrepancies between injections (random error)

Curve of ratios of peak area (sample:IS) plotted vs [Sample]:[IS]



Internal Standard

Benzene (0.1 mg/ml)	Toluene (0.1 mg/ml)	Ratio (PA_B/PA_T)
105	105	1
115	115	1
97.0	97.0	1

Differences in repeat benzene measurements
are seen in the internal standard toluene

Internal Standard

$$\frac{A_X}{[X]_f} = F \left(\frac{A_S}{[S]_f} \right)$$

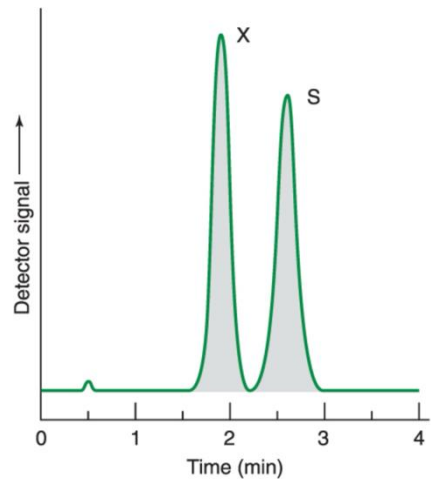
A_X = signal from analyte
 $[X]_f$ = concentration of analyte
 A_S = signal from standard
 $[S]_f$ = concentration of standard
 F = response factor

Use a standard and internal standard of known concentration to calculate the response factor, F .

Use F to calculate the concentration of your sample with the internal standard at the same concentration.

Practice Problem

In a preliminary experiment, a solution containing 0.0837 M X and 0.0666 M S, gave peak areas of $A_x=432$ and $A_s=347$ (Areas are measured in arbitrary units by the instrument.) To analyze the unknown, 10.0 mL of 0.146 M S were added to 10.0 mL of unknown, and the mixture was diluted to 25.0 mL in a volumetric flask. This mixture gave the below chromatogram, for which $A_x=553$ and $A_s=582$. Find the concentration of X in the unknown.



Internal Standard

Internal standards are especially useful for analyses in which the quantity of sample analyzed or the instrument response varies slightly from run to run.

What would be some good characteristics of an internal standard?

- Well-resolved from analyte

- Stable

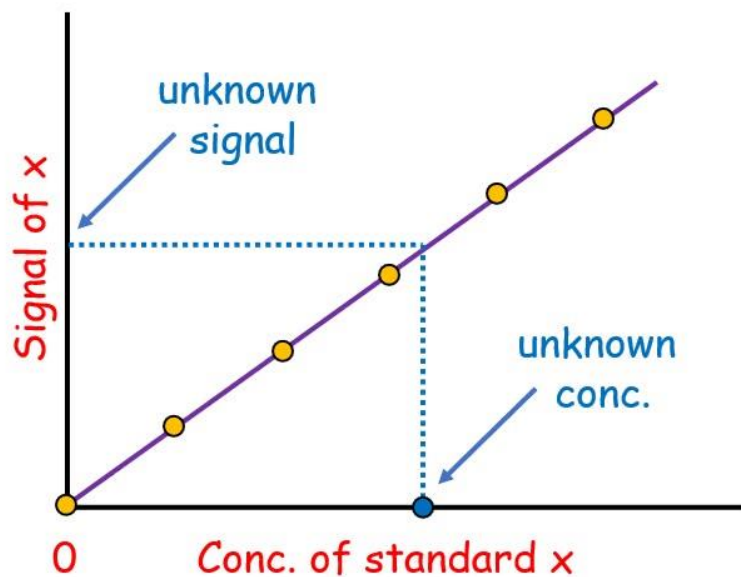
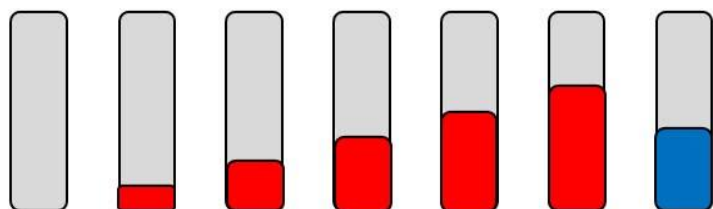
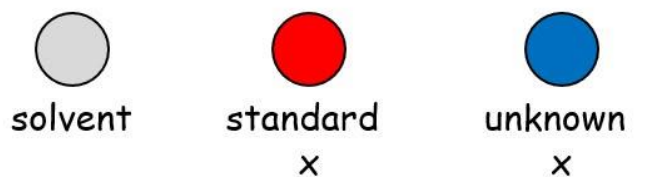
- Pure

- High response from detector

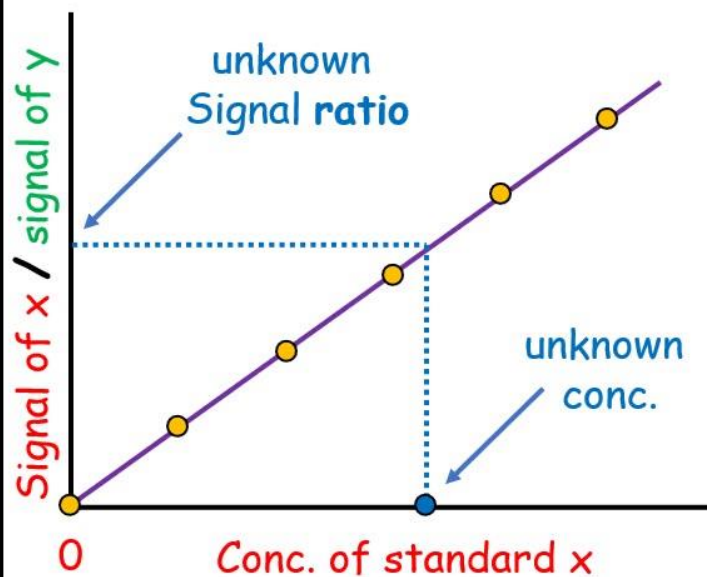
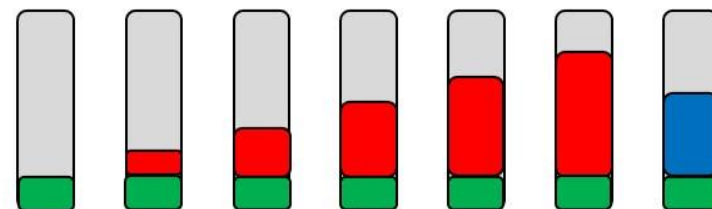
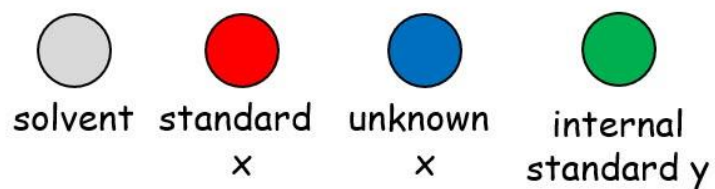
- Never found in sample

- Structurally similar to analyte

External standard



Internal standard



Standard addition

